THREE-STATE GATED TWO-HARMONIC DISTRIBUTIONS: CRITICAL-POINT CLASSIFICATION, STATISTICAL INFERENCE, PHOTONIC BANDS, AND AN ARITHMETIC EXAMPLE

Abstract. We study a gated family obtained by pushing uniform circle measure through two-frequency trigonometric polynomials. We prove an inverse-branch density formula, an absolutely convergent resonant Bessel expansion, exact moment selection rules, a symmetry classification, and a complete classification of multiple critical points for distinct frequencies. In particular, critical orders are at most four. For i.i.d. observations in a compact separated parameter chamber, we construct moment and empirical- characteristic-function estimators, prove consistency and asymptotic normality, derive the ECF sandwich covariance, establish parametric-bootstrap validity, and give exact or conservative finite-sample confidence sets by distribution-function inversion. Unknown frequencies are treated separately as finite discrete model selection under an explicit cross-model gap. We derive a photonic tight-binding realization and identify a tunable cubic transition in a published synthetic-frequency platform. As a separate arithmetic example, Möbius sampling with phase $\omega k \log k$ satisfies unconditional quantitative Kolmogorov bounds, including an effective rate of order $(\log\log N)(\log N)^{-1/4}$; random-index inference for a fixed finite prefix is handled by hypergeometric inversion and Serfling bounds, without interpreting those design intervals as rates for the infinite arithmetic limit.

Contents

1. Introduction 2
2. Probability law 3
3. Characteristic function and moments 3
4. Symmetry and identifiability 4
5. Critical values 5
5.1. The one-dimensional Van Hove mechanism 5
6. Limits 7
7. An arithmetic example: Möbius sampling 8
7.1. Quantitative Kolmogorov discrepancy 9
8. Statistical inference 12
8.1. Regular parameter chambers 12
8.2. Moment and ECF estimators 12
8.3. Asymptotic normality and sandwich covariance 14
8.4. Finite-sample confidence sets 16
8.5. Parametric bootstrap 17
8.6. Unknown frequencies as discrete model selection 18
8.7. Random-index inference for a finite Möbius population 18
8.8. Simulation audit 20
9. Numerical audit of the arithmetic example 21
10. Photonic realization: the Dutt et al. experiment 22
10.1. Experimental input and scope 22
10.2. Exact map into the harmonic family 23
10.3. Calibration, broadening, and active-fraction recovery 23
10.4. A nearby, testable singularity transition 24
10.5. What the generalization adds 26
11. Homogeneous-domain interpretation 27

References 27

1. Introduction

The experimental work of Dutt et al. [9] provides the physical motivation for the present study. That work introduced direct band-structure spectroscopy in a synthetic frequency dimension and demonstrated reconfigurable long-range hopping, a photonic gauge phase, asymmetric bands, and direct access to the density of states and its Van Hove singularities. In particular, its two-drive dispersion contains a first cosine harmonic and a phase-shifted higher harmonic. At gauge phase $\pi/2$, this is, up to an energy scale and reflection of quasimomentum, precisely a function of the form

$$\lambda \sin(r\theta) + \eta \cos(s\theta). \tag{1.1}$$

The relevance and experimental flexibility of this platform motivated us to ask for a systematic probabilistic generalization of such dispersions: their complete value laws under uniform quasimomentum, the singularities of those laws, the exact information contained in their moments and characteristic functions, and the effect of mixing distinct homogeneous gauge domains.

The present paper studies the first nontrivial gated version of this program. Let Θ be uniform on $[0, 2\pi)$ and independent of M, where

$$P(M = 0) = 1 - q, \qquad P(M = 1) = P(M = -1) = q/2. \tag{1.2}$$

For positive λ, η and coprime $r, s \in \mathbb{N}$, define

$$X = \lambda M \sin(r\Theta) + \eta \cos(s\Theta). \tag{1.3}$$

Thus the model already has three gate states: two time-reversed active domains and one inactive higher-harmonic domain. The two-state model is the restriction $M \in \{-1, 1\}$, equivalently $q = 1$. A later extension may replace M by an arbitrary finite categorical gate, or allow several independently gated harmonics. We do not develop that larger model here, because its identifiability and degeneracy geometry require a separate analysis.

As one example, the Möbius function supplies a deterministic sampling mechanism for this probability law, but it does not define or restrict the probabilistic family. It is a particularly natural three-state gate because

$$\mu(k) \in \{-1, 0, 1\}, \qquad \operatorname{dens}\{\mu = 0\} = 1 - \frac{6}{\pi^2}, \qquad \operatorname{dens}\{\mu = \pm 1\} = \frac{3}{\pi^2}. \tag{1.4}$$

The zero state records nonsquarefree integers, while the two active states record the parity of the number of prime factors on the squarefree integers. Thus Möbius supplies the same gate weights as (1.2) with $q = 6/\pi^2$, together with substantial known multiplicative and equidistribution structure. The gate and phase are not assumed independent at finite sample size; the arithmetic theorem proves that the corresponding mixture nevertheless emerges asymptotically.

The circle-pushforward theory and its physical DOS interpretation remain meaningful for laboratory quasimomentum samples, random categorical gates, other arithmetic weights, and deterministic sequences satisfying appropriate joint equidistribution conditions. The four layers of the paper are logically separate. Sections 2– 6 concern the exact circle pushforward; Section 8 assumes an explicitly i.i.d. sample; Sections 7 and 9 concern a deterministic arithmetic sequence and use no i.i.d. calibration; Sections 10–11 concern homogeneous Bloch bands and their uniform-quasimomentum DOS. No conclusion is transferred between these four settings without a separately stated theorem. Circle pushforwards and density-of-states singularities are standard; see [12, 2]. Our contribution is the integrated analysis of (1.3), including an explicit finite degeneracy set for every coprime frequency pair and a proof that no critical order above four can occur. Möbius sampling is treated

only as a subsequent arithmetic example. The literature audit accompanying this paper does not justify claiming that trigonometric pushforwards or higher-order Van Hove singularities themselves are new. A subscriber MathSciNet audit and a complete Google Scholar exact-formula search remain pending.

2. Probability law

Put $h(\theta) = \lambda \sin(r\theta) + \eta \cos(s\theta)$.

Lemma 2.1. *The conditional laws for $M = 1$ and $M = -1$ coincide. Hence*

$$\mathcal{L}(X) = (1-q)(\eta \cos(s\Theta))_{\#}m + q(h(\Theta))_{\#}m, \tag{2.1}$$

where m is normalized circle measure.

Proof. Use $\theta \mapsto -\theta$. □

Theorem 2.2. *For every*

$$x \notin \mathrm{CritVal}(h) \cup \{-\eta, \eta\}, \tag{2.2}$$

the density is given pointwise by

$$f_X(x) = \frac{(1-q)\mathbf{1}_{|x|<\eta}}{\pi\sqrt{\eta^2 - x^2}} + \frac{q}{2\pi} \sum_{h(\theta)=x} \frac{1}{|h'(\theta)|}. \tag{2.3}$$

If $q > 0$, then

$$\mathrm{supp}(X) = [\min h, \max h]. \tag{2.4}$$

If $q = 0$, then

$$\mathrm{supp}(X) = [-\eta, \eta]. \tag{2.5}$$

Proof. Partition the circle at the finitely many critical phases of h. On each resulting monotonicity interval, the change-of-variables formula gives the active term in (2.3); the arcsine change of variables gives the first term. These formulas are pointwise valid away from the active critical values and the arcsine edges.

The image of the circle under h is the interval $[\min h, \max h]$. If $\cos(s\theta) = \pm 1$, then

$$\frac{h(\theta) + h(-\theta)}{2} = \eta \cos(s\theta) = \pm\eta. \tag{2.6}$$

Because the image is an interval, it contains both $-\eta$ and η, and hence it contains $[-\eta, \eta]$. Therefore, when $q > 0$, the support of the mixture is the active support $[\min h, \max h]$. When $q = 0$, only the arcsine component remains, whose support is $[-\eta, \eta]$. □

3. Characteristic function and moments

Theorem 3.1. *For every real t,*

$$\phi_X(t) = (1-q)J_0(\eta t) + q \sum_{\ell \in \mathbb{Z}} i^{r\ell} J_{s\ell}(\lambda t) J_{r\ell}(\eta t), \tag{3.1}$$

and the series converges absolutely and locally uniformly.

Proof. Apply the Jacobi–Anger expansions [8]; circle integration retains precisely $mr + ns = 0$, hence $(m, n) = (s\ell, -r\ell)$. The standard factorial bounds for integer-order Bessel functions make both Jacobi–Anger series absolutely and uniformly convergent when t ranges over a compact set. Explicitly, for $n \geq 0$ and $|x| \leq T$, the defining series and $(n+j)! \geq n!j!$ give

$$|J_n(x)| \leq \frac{(T/2)^n}{n!} \sum_{j \geq 0} \frac{(T^2/4)^j}{(j!)^2} \leq \frac{(T/2)^n}{n!} e^{T^2/4}, \qquad |J_{-n}(x)| = |J_n(x)|. \tag{3.2}$$

The Weierstrass test therefore applies to both Jacobi–Anger series and their product. Fubini's theorem permits termwise circle integration, and the retained one-dimensional subseries is absolutely and locally uniformly convergent. □

For $a, b \geq 0$, set $I_{a,b} = \int \sin^a(r\theta)\cos^b(s\theta)\, dm$.

Proposition 3.2.

$$I_{a,b} = \frac{1}{(2i)^a 2^b} \sum_{j=0}^{a} \sum_{k=0}^{b} (-1)^j \binom{a}{j} \binom{b}{k} \mathbf{1}_{r(a-2j)+s(b-2k)=0}. \tag{3.3}$$

Thus all moments and cumulants are finite explicit sums. If $r \neq s$,

$$E[X^2] = (\eta^2 + q\lambda^2)/2, \tag{3.4}$$

$$E[X^4] = 3\eta^4/8 + q(3\lambda^4/8 + 3\lambda^2\eta^2/2). \tag{3.5}$$

Proof. The exponential identities

$$\sin^a(r\theta) = \frac{1}{(2i)^a} \sum_{j=0}^{a} (-1)^j \binom{a}{j} e^{ir(a-2j)\theta}, \qquad \cos^b(s\theta) = \frac{1}{2^b} \sum_{k=0}^{b} \binom{b}{k} e^{is(b-2k)\theta} \tag{3.6}$$

give (3.3) after integration, because only the zero Fourier frequency survives. Expanding powers of (1.3) and using $E[M^j] = 0$ for odd j and $E[M^j] = q$ for positive even j reduces every moment to these finite sums. Finally,

$$I_{2,0} = I_{0,2} = \frac{1}{2}, \qquad I_{4,0} = I_{0,4} = \frac{3}{8}, \qquad I_{2,2} = \frac{1}{4} \quad (r \neq s), \tag{3.7}$$

which gives (3.4) and (3.5). □

4. Symmetry and identifiability

Theorem 4.1. *For $q\lambda\eta > 0$ and reduced frequencies, X is symmetric if and only if s is odd. If s is even, then r is odd and*

$$E[X^{r+s}] = q \binom{r+s}{s} 2^{1-r-s} (-1)^{s/2} \lambda^s \eta^r \neq 0. \tag{4.1}$$

Proof. Suppose first that s is odd. If r is odd, the measure-preserving map $\theta \mapsto \theta + \pi$ changes the signs of both harmonics. If r is even, the map $\theta \mapsto \pi - \theta$ does so. Hence the law is symmetric.

Now let s be even. Coprimality forces r to be odd, so $n = r+s$ is odd. In the binomial expansion of $E[X^n]$, terms with an odd power a of the sine harmonic vanish because $E[M^a] = 0$, and the term $a = 0$ is the mean of an odd power of a cosine. For positive even a put $b = n - a$. A nonzero Fourier contribution to $I_{a,b}$ requires integers j, k such that

$$r(a - 2j) + s(b - 2k) = 0. \tag{4.2}$$

Coprimality gives $a - 2j = sL$ and $b - 2k = -rL$. Here L is odd, while $s|L| \leq a$, $r|L| \leq b$, and $a + b = r + s$; consequently $L = \pm 1$ and necessarily $(a, b) = (s, r)$. The two retained Fourier terms in (3.3) give

$$I_{s,r} = 2^{1-r-s}(-1)^{s/2}. \tag{4.3}$$

Multiplication by the binomial coefficient, amplitudes, and $E[M^s] = q$ proves (4.1). This nonzero odd moment excludes symmetry. □

Proposition 4.2. *Fix distinct positive integers r and s, and let*

$$\mathcal{P} = (0,\infty)^2 \times (0,1) \tag{4.4}$$

be the parameter space for $p = (\lambda, \eta, q)$. For $h_p(\theta) = \lambda \sin(r\theta) + \eta \cos(s\theta)$, fix $p_0 = (\lambda_0, \eta_0, q_0) \in \mathcal{P}$. Suppose that there exist an open neighborhood $U \subset \mathcal{P}$ of p_0 and a number $\delta > 0$ such that

$$\operatorname{dist}(\{-\eta, \eta\}, \operatorname{CritVal}(h_p)) > \delta \qquad \text{for every } p = (\lambda, \eta, q) \in U. \tag{4.5}$$

Then $p \mapsto \mathcal{L}(X_p)$ is injective on a possibly smaller neighborhood of p_0. In particular, (λ, η, q) is locally identifiable. Once η has been recovered from the law, put

$$A = 2E[X^2] - \eta^2, \qquad B = \frac{8}{3}\left(E[X^4] - \frac{3\eta^4}{8}\right). \tag{4.6}$$

Then

$$\lambda^2 = B/A - 4\eta^2, \qquad q = A/\lambda^2. \tag{4.7}$$

Proof. Because $0 < q < 1$, the arcsine component has positive mass. By (4.5), the active density has no critical-value singularity in the δ-neighborhood of either arcsine edge. Hence the law has two isolated square-root singularities at the symmetric locations $-\eta$ and η that persist throughout U. Their locations determine the positive parameter η.

The second and fourth moments are observables of the law. Equations (3.4) and (3.5) give

$$A = q\lambda^2, \qquad B = q\lambda^4 + 4q\lambda^2\eta^2 = A(\lambda^2 + 4\eta^2). \tag{4.8}$$

Since $q > 0$ and $\lambda > 0$, one has $A > 0$. Therefore the law determines $\lambda^2 = B/A - 4\eta^2$, and positivity selects the unique value $\lambda > 0$. It then determines $q = A/\lambda^2$. Thus any two parameters in a sufficiently small neighborhood of p_0 with the same law have identical η, λ, and q. □

Degeneracies include $q = 0$, $\lambda = 0$, and $r = s$, the latter reducing to an arcsine component of radius $\sqrt{\lambda^2 + \eta^2}$.

5. Critical values

5.1. **The one-dimensional Van Hove mechanism.** For a one-dimensional band $\varepsilon(k)$ sampled with uniform quasimomentum, the density of states is the pushforward density

$$g(E) = \frac{1}{2\pi} \sum_{\varepsilon(k)=E} \frac{1}{|\varepsilon'(k)|}. \tag{5.1}$$

It is smooth at regular energies but becomes singular at a critical energy $E_c = \varepsilon(k_c)$ where $\varepsilon'(k_c) = 0$. At an ordinary quadratic band edge, $\varepsilon(k) - E_c$ is proportional to $(k - k_c)^2$, and hence $g(E)$ diverges on one side as $|E - E_c|^{-1/2}$. If the first nonzero derivative has order $m > 2$, the band is locally flatter and the divergence strengthens to $|E - E_c|^{1/m-1}$. This critical-point mechanism goes back to Van Hove [12]; higher-order versions are now standard objects of band theory [2]. Our new question is not whether such singularities exist, but exactly which orders and tuning ratios are possible in the constrained two-frequency family (5.6).

Theorem 5.1. *Assume $q > 0$. If $h^{(1)}(\theta_0) = \cdots = h^{(m-1)}(\theta_0) = 0$ and $h^{(m)}(\theta_0) \neq 0$, put $x_0 = h(\theta_0)$ and $a_m = h^{(m)}(\theta_0)/m!$. On every locally accessible side of x_0, the contribution of this phase to the active-branch density satisfies*

$$f_{\theta_0}(x) \sim \frac{qN_m}{2\pi m |a_m|^{1/m}} |x - x_0|^{1/m-1}, \tag{5.2}$$

where $N_m = 1$ for odd m and $N_m = 2$ for even m. In the even case the accessible side is $a_m(x-x_0) > 0$, and the phase contributes nothing on the other side sufficiently close to x_0. In particular, the exponent is

$$|x - h(\theta_0)|^{1/m-1}. \tag{5.3}$$

For even m it is one-sided; for odd m it occurs on both sides.

Proof. Write $u = \theta - \theta_0$. Taylor's theorem gives

$$h(\theta_0 + u) - x_0 = a_m u^m\{1 + o(1)\}, \qquad h'(\theta_0 + u) = m a_m u^{m-1}\{1 + o(1)\}. \tag{5.4}$$

Along each accessible inverse branch,

$$|u| = |a_m|^{-1/m}|x - x_0|^{1/m}\{1 + o(1)\}, \quad \frac{1}{|h'(\theta_0 + u)|} = \frac{|x - x_0|^{1/m-1}}{m|a_m|^{1/m}}\{1 + o(1)\}. \tag{5.5}$$

An odd power has one local real inverse branch on each side; an even power has two on the side $a_m(x - x_0) > 0$ and none on the other side. Multiplication by the active mass $q/(2\pi)$ and the corresponding branch count N_m proves (5.2). □

If $q = 0$, the active component in (2.1) has mass zero and contributes no density.

Theorem 5.2 (Complete multiple-critical-point classification)**.** *Let r and s be distinct coprime positive integers and let $\lambda, \eta > 0$. Every critical point of*

$$h(\theta) = \lambda \sin(r\theta) + \eta \cos(s\theta) \tag{5.6}$$

has order 2, 3, or 4. All points of order at least 3 are obtained as follows.

Away from the quartic case below, their phases are precisely the solutions of

$$(r + s)\cos((r - s)\theta) = (r - s)\cos((r + s)\theta) \tag{5.7}$$

for which

$$\frac{\lambda}{\eta} = \frac{s \sin(s\theta)}{r \cos(r\theta)} > 0. \tag{5.8}$$

Every such point has order 3 and density exponent $-2/3$.

An order-4 point exists if and only if

$$r \text{ is odd}, \qquad s \text{ is even}, \qquad \frac{\lambda}{\eta} = \frac{s^2}{r^2}. \tag{5.9}$$

For positive amplitudes exactly one of $\theta = \pi/2$ and $\theta = 3\pi/2$ is the quartic phase. Its density exponent is $-3/4$. Consequently, for each fixed (r, s) the set of amplitude ratios at which the number of critical points can change is finite and explicitly computable from (5.7) *and* (5.9)*.*

Proof. The first four derivatives are

$$h' = \lambda r \cos(r\theta) - \eta s \sin(s\theta), \tag{5.10}$$

$$h'' = -\lambda r^2 \sin(r\theta) - \eta s^2 \cos(s\theta), \tag{5.11}$$

$$h''' = -\lambda r^3 \cos(r\theta) + \eta s^3 \sin(s\theta), \tag{5.12}$$

$$h^{(4)} = \lambda r^4 \sin(r\theta) + \eta s^4 \cos(s\theta). \tag{5.13}$$

The determinant of the homogeneous linear system in (λ, η) given by $h' = h'' = 0$ vanishes precisely when

$$s \cos(r\theta)\cos(s\theta) + r \sin(r\theta)\sin(s\theta) = 0. \tag{5.14}$$

The product-to-sum identities turn (5.14) into (5.7). If $\cos(r\theta) \neq 0$, the first derivative then gives (5.8). Conversely, a phase satisfying these two equations makes both (5.10) and (5.11) vanish.

At any critical point, set $C = \lambda r\cos(r\theta) = \eta s\sin(s\theta)$. Equation (5.12) gives

$$h'''(\theta) = (s^2 - r^2)C. \tag{5.15}$$

Because $r \neq s$, a multiple critical point is cubic unless $C = 0$. In the latter case, positivity of the amplitudes implies

$$\cos(r\theta) = 0, \qquad \sin(s\theta) = 0. \tag{5.16}$$

For coprime frequencies, (5.16) is solvable precisely when s is even; then r is odd and the only phases modulo 2π are $\pi/2$ and $3\pi/2$. At these phases, $h'' = 0$ has a positive-amplitude solution at exactly one of the two phases. Indeed,

$$\frac{\lambda}{\eta} = -\frac{s^2\cos(s\theta)}{r^2\sin(r\theta)}. \tag{5.17}$$

The cosine has the same sign at the two candidate phases, whereas the sine has opposite signs. Exactly one ratio is therefore positive, and its modulus is s^2/r^2.

Finally, using $h'' = 0$ in (5.13) gives

$$h^{(4)}(\theta) = \lambda r^2(r^2 - s^2)\sin(r\theta) \neq 0. \tag{5.18}$$

Thus the remaining point is quartic and no order above four is possible. The left-hand side of (5.7) is a nonzero trigonometric polynomial of degree $r + s$, so it has only finitely many phases. Away from their associated ratios, the implicit-function theorem preserves the number of critical points. □

General higher-order Van Hove classification and catastrophe-theory normal forms are established topics [2]. Theorem 5.2 is instead an exact global classification within the constrained two-frequency family. Its novelty status remains subject to the source-by-source audit accompanying this paper.

For $(r, s) = (2, 1)$ and $\rho = \eta/\lambda$, stationary points satisfy

$$u_\pm = \frac{-\rho \pm \sqrt{\rho^2 + 32}}{8}, \qquad c_\pm = \lambda\sqrt{1 - u_\pm^2}(2u_\pm + \rho). \tag{5.19}$$

The u_- branch exists for $\rho \leq 2$; at $\rho = 2$ it is cubic and gives exponent $-2/3$. For $\rho = 1/4$ the six singular values are ± 1.1805175473, ± 0.8273102715, and ± 0.25.

6. Limits

Scaling (λ, η) scales X. As $q \to 0$ or $\lambda \to 0$ one obtains the arcsine law of radius η. As $\eta \to 0$, an atom of mass $1 - q$ at zero remains.

Proposition 6.1. *If reduced pairs (r_n, s_n) escape every finite set, the circle measures $\theta \mapsto (r_n\theta, s_n\theta)$ converge weakly to Haar measure on the two-torus; the branch limit is therefore the convolution of independent arcsine laws.*

Proof. For $(u, v) \in \mathbb{Z}^2$, the (u, v) Fourier coefficient of the image measure is

$$\frac{1}{2\pi}\int_0^{2\pi} e^{i(ur_n + vs_n)\theta}\, d\theta = \mathbf{1}_{\{ur_n + vs_n = 0\}}. \tag{6.1}$$

For fixed $(u, v) \neq (0, 0)$, the last equality can hold for at most one coprime positive pair (r_n, s_n). Hence every nonconstant Fourier coefficient tends to zero, which is the Fourier characterization of Haar convergence on the two-torus. Applying the continuous map $(x, y) \mapsto \lambda\sin x + \eta\cos y$ gives the stated convolution. □

7. An arithmetic example: Möbius sampling

The role of Möbius in this section is illustrative but nontrivial. Unlike an artificial i.i.d. categorical draw, it is a deterministic three-state sequence whose state frequencies are known and whose signed squarefree structure is arithmetic. It therefore tests whether the same gated law can arise from structured deterministic sampling rather than from the independence assumption used in (1.2). Define

$$F(k) = \lambda\mu(k)\sin(r\omega k\log k) + \eta\cos(s\omega k\log k). \tag{7.1}$$

Theorem 7.1. *For every fixed $\omega \neq 0$, the empirical measures $N^{-1}\sum_{k\le N}\delta_{F(k)}$ converge weakly to (2.1) with $q = 6/\pi^2$.*

Proof. For $z \in \{-1, 0, 1\}$, let

$$E_z = \{n \in \mathbb{N} : \mu(n) = z\}. \tag{7.2}$$

Since the Möbius function is multiplicative, each E_z is a one-function level set and therefore belongs to the class $\mathcal{D}^{(1)}$ in Definition 1.3 of [1]. The squarefree-density theorem gives

$$d(E_{-1}) + d(E_1) = \frac{6}{\pi^2}, \qquad d(E_0) = 1 - \frac{6}{\pi^2}. \tag{7.3}$$

The prime number theorem is equivalent to $N^{-1}\sum_{n\le N}\mu(n) \to 0$, and hence

$$d(E_{-1}) = d(E_1) = \frac{3}{\pi^2}. \tag{7.4}$$

Thus every E_z has an existing positive density. Proposition 3.2 on p. 2591 of the cited paper then places each E_z in $\mathcal{E}^{(1)}_{\text{c.pt.}}$, exactly one of the two set classes admitted by its Theorem 4.1 on pp. 2607–2608.

Set

$$h_\omega(t) = \frac{\omega}{2\pi}t\log t. \tag{7.5}$$

For $\omega \neq 0$, this function belongs to the logarithmico-exponential Hardy field. It has polynomial growth because $|h_\omega(t)|/t^2 \to 0$. Let $P \in \mathbb{Q}[t]$. If $\deg P \ge 2$, then $|h_\omega(t) - P(t)|$ is asymptotic to the modulus of the leading term of P. If $\deg P \le 1$, then

$$|h_\omega(t) - P(t)| \sim \frac{|\omega|}{2\pi}t\log t. \tag{7.6}$$

In both cases,

$$\frac{|h_\omega(t) - P(t)|}{\log^2 t} \longrightarrow \infty. \tag{7.7}$$

All hypotheses of Theorem 4.1 of [1] therefore hold for each of the three sets E_z. If $E_z = \{n_{z,1} < n_{z,2} < \cdots\}$, the sequence $h_\omega(n_{z,j})$ is uniformly distributed modulo one. Equivalently, $\omega n_{z,j}\log n_{z,j}$ is uniformly distributed modulo 2π.

Let ψ be bounded and continuous, and put

$$H_z(\theta) = \psi\big(\lambda z\sin(r\theta) + \eta\cos(s\theta)\big). \tag{7.8}$$

Uniform distribution on E_z and the density of E_z give

$$\frac{1}{N}\sum_{\substack{k\le N\\ k\in E_z}}\psi(F(k)) \longrightarrow d(E_z)\frac{1}{2\pi}\int_0^{2\pi} H_z(\theta)\,d\theta. \tag{7.9}$$

Summing (7.9) over $z = -1, 0, 1$, using (7.4), and applying Lemma 2.1 yields

$$\frac{1}{N}\sum_{k\le N}\psi(F(k)) \longrightarrow \left(1-\frac{6}{\pi^2}\right)\int \psi\, d(\eta\cos(s\Theta))_{\#}m + \frac{6}{\pi^2}\int \psi\, d(h(\Theta))_{\#}m. \tag{7.10}$$

This is the asserted weak convergence. □

7.1. **Quantitative Kolmogorov discrepancy.** The qualitative theorem admits an unconditional quantitative strengthening. Define

$$\mathcal{F}_\infty(x) = \left(1-\frac{6}{\pi^2}\right)\frac{1}{2\pi}\,\mathrm{meas}\{\theta : \eta\cos(s\theta)\le x\} + \frac{6}{\pi^2}\frac{1}{2\pi}\,\mathrm{meas}\{\theta : h(\theta)\le x\} \tag{7.11}$$

and

$$\mathfrak{D}_N = \sup_{x\in\mathbb{R}}\left|\frac{1}{N}\#\{k\le N : F(k)\le x\} - \mathcal{F}_\infty(x)\right|. \tag{7.12}$$

Theorem 7.2 (Quantitative arithmetic discrepancy)**.** *Fix $\lambda, \eta > 0$, positive integers r, s, and $\omega \neq 0$. For every $A > 0$,*

$$\mathfrak{D}_N \ll_{A,\lambda,\eta,r,s,\omega} (\log N)^{-A}, \tag{7.13}$$

where the implied constant is ineffective. One also has the effective estimate

$$\mathfrak{D}_N \ll_{\lambda,\eta,r,s,\omega} \frac{\log\log N}{(\log N)^{1/4}}. \tag{7.14}$$

Both estimates hold for all sufficiently large N and use no unproved number-theoretic hypothesis. Effectivity in (7.14) *means that the implied constant and threshold are computable in principle; no numerical value for either is asserted here.*

Write $e(t) = \exp(2\pi i t)$ and, for $a \in \mathbb{R}$, set

$$U_a(N) = \sum_{n\le N} e(an\log n), \tag{7.15}$$

$$Q_a(N) = \sum_{n\le N} \mu(n)^2 e(an\log n), \tag{7.16}$$

$$M_a(N) = \sum_{n\le N} \mu(n) e(an\log n). \tag{7.17}$$

Lemma 7.3. *For $a \neq 0$,*

$$U_a(N) \ll \left(|a|^{1/2} + |a|^{-1/2}\right)N^{1/2} + 1. \tag{7.18}$$

Proof. On $X < n \le 2X$, the second derivative of $at\log t$ has modulus between $|a|/(2X)$ and $|a|/X$. The van der Corput second-derivative estimate [5, Chapter 2] gives

$$\sum_{X<n\le 2X} e(an\log n) \ll |a|^{1/2}X^{1/2} + |a|^{-1/2}X^{1/2}. \tag{7.19}$$

Summation over dyadic intervals proves (7.18). □

Lemma 7.4. *For $a \neq 0$ and N sufficiently large in terms of a,*

$$Q_a(N) \ll |a|^{1/4}N^{3/4} + |a|^{-1/2}N^{1/2} + N^{1/2}. \tag{7.20}$$

For every fixed $c, C > 0$, the estimate is uniform when $c \le |a| \le (\log N)^C$.

Proof. Use

$$\mu(n)^2 = \sum_{d^2 \mid n} \mu(d). \tag{7.21}$$

Split the resulting d-sum at $1 \le Y \le N^{1/2}$. For $d \le Y$, apply the second-derivative estimate to

$$g_d(t) = ad^2 t \log(d^2 t), \qquad g_d''(t) = \frac{ad^2}{t}. \tag{7.22}$$

After dyadic subdivision,

$$\sum_{m \le N/d^2} e(ad^2 m \log(d^2 m)) \ll |a|^{1/2} N^{1/2} + \frac{N^{1/2}}{|a|^{1/2} d^2} + 1. \tag{7.23}$$

The range $d > Y$ contributes at most $O(N/Y)$. Consequently,

$$Q_a(N) \ll Y|a|^{1/2} N^{1/2} + |a|^{-1/2} N^{1/2} + Y + \frac{N}{Y}. \tag{7.24}$$

Taking Y to be the nearest admissible integer to $|a|^{-1/4} N^{1/4}$ proves the result. □

Proposition 7.5. *Fix $B > 0$. Uniformly for $|a| \le (\log N)^B$,*

$$M_a(N) \ll_B N(\log N)^{-1/4} + (1 + |a|) N^{2/3}, \tag{7.25}$$

with an effective implied constant. For every $A > 0$,

$$M_a(N) \ll_{A,B} N(\log N)^{-A} + (1 + |a|) N^{2/3}, \tag{7.26}$$

where the implied constant is ineffective.

Proof. We use Theorem 1.1(i) and (iv) of Matomäki, Shao, Tao, and Teräväinen [7]. Their theorem is uniform over polynomial phases of fixed degree on intervals $(X, X + H]$ with $H \ge X^{5/8+\varepsilon}$ for an arbitrary logarithmic saving and with $H \ge X^{3/5+\varepsilon}$ for the effective saving $(\log X)^{-1/4}$.

Partition $(X, 2X]$ into full intervals $I = (y, y + H]$ with $H = \lfloor X^{2/3} \rfloor$. Keep a possible final interval of length less than H separate and bound it trivially. On a full interval let P_y be the cubic Taylor polynomial of $at \log t$ at y. Since

$$\frac{d^4}{dt^4}(at \log t) = \frac{2a}{t^3}, \tag{7.27}$$

Taylor's theorem gives

$$\sup_{t \in I} |at \log t - P_y(t)| \ll |a| \frac{H^4}{X^3} \ll |a| X^{-1/3}. \tag{7.28}$$

Replacing the phase by P_y therefore costs $O(|a| H X^{-1/3})$ on I.

Choose a fixed $0 < \varepsilon < 1/24$. Uniformly for $X \le y \le 2X$ and all sufficiently large X,

$$y^{5/8+\varepsilon} \le H \le y^{1-\varepsilon}. \tag{7.29}$$

This verifies the hypotheses of both cited parts. Polynomial phases of degree at most three arise from the fixed filtered circle $\mathbb{R}/\mathbb{Z}$ and the fixed Lipschitz function $t \mapsto e(t)$, so all nilmanifold parameters are independent of a, X, and y. The source theorem is also stated for maximal sums over arithmetic progressions and hence covers the ordinary interval sums used here. Summing over the $O(X/H)$ full blocks, adding the final $O(H)$ block, and using (7.28) gives

$$\sum_{X < n \le 2X} \mu(n) e(an \log n) \ll_A X(\log X)^{-A} + (1 + |a|) X^{2/3}, \tag{7.30}$$

$$\sum_{X < n \le 2X} \mu(n) e(an \log n) \ll X(\log X)^{-1/4} + (1 + |a|) X^{2/3}. \tag{7.31}$$

The first constant is ineffective and the second is effective, exactly as stated after Theorem 1.1 of the cited paper. Dyadically sum $N^{1/2} < n \le N$ and bound the initial range trivially. The restriction $|a| \le (\log N)^B$ absorbs the initial and Taylor errors into (7.25) and (7.26). □

Proof of Theorem 7.2. Put

$$u_n = \frac{\omega}{2\pi} n \log n \pmod 1 \tag{7.32}$$

and, for $z \in \{-1, 0, 1\}$, define

$$S_z(m; N) = \sum_{n \le N} \mathbf{1}_{\{\mu(n)=z\}} e(m u_n). \tag{7.33}$$

The identities

$$\mathbf{1}_{\{\mu=0\}} = 1 - \mu^2, \tag{7.34}$$

$$\mathbf{1}_{\{\mu=1\}} = (\mu^2 + \mu)/2, \tag{7.35}$$

$$\mathbf{1}_{\{\mu=-1\}} = (\mu^2 - \mu)/2 \tag{7.36}$$

reduce every $S_z(m; N)$ with $m \neq 0$ to a fixed linear combination of $U_a(N)$, $Q_a(N)$, and $M_a(N)$, where

$$a = \frac{m\omega}{2\pi}. \tag{7.37}$$

Let $d_0 = 1 - 6/\pi^2$ and $d_1 = d_{-1} = 3/\pi^2$. For a circle interval I, set

$$\Delta_z(N) = \sup_I \left| \frac{1}{N} \sum_{n \le N} \mathbf{1}_{\{\mu(n)=z\}} \mathbf{1}_I(u_n) - d_z |I| \right|. \tag{7.38}$$

At frequency zero,

$$\sum_{n \le N} \mu(n)^2 = \frac{6}{\pi^2} N + O(N^{1/2}), \tag{7.39}$$

while Proposition 7.5 with $a = 0$ controls the signed term. The weighted Erdős–Turán inequality [6, Chapter 2] therefore gives, for every $L \ge 1$,

$$\Delta_z(N) \ll \frac{|S_z(0; N) - d_z N|}{N} + \frac{1}{L} + \sum_{1 \le m \le L} \frac{|S_z(m; N)|}{mN}. \tag{7.40}$$

With $L = \lfloor (\log N)^{1/4} \rfloor$, the preceding estimates yield

$$\sum_{1 \le m \le L} \frac{|S_z(m; N)|}{mN} \ll_\omega \frac{\log L}{(\log N)^{1/4}} + \frac{L}{N^{1/3}} + \frac{L^{1/4}}{N^{1/4}} + \frac{L^{1/2}}{N^{1/2}}. \tag{7.41}$$

This proves

$$\Delta_z(N) \ll_\omega \frac{\log \log N}{(\log N)^{1/4}}. \tag{7.42}$$

For prescribed $A > 0$, take $L = \lfloor (\log N)^{A+1} \rfloor$ and apply (7.26) with a logarithmic exponent larger than $2A + 3$. The same calculation gives

$$\Delta_z(N) \ll_{A,\omega} (\log N)^{-A} \tag{7.43}$$

with an ineffective constant.

For fixed z and x, the set

$$\mathcal{A}_z(x) = \{u \in \mathbb{R}/\mathbb{Z} : \lambda z \sin(2\pi r u) + \eta \cos(2\pi s u) \le x\} \tag{7.44}$$

is a union of at most $2\max\{r,s\}$ intervals: its boundary lies in the zero set of a nonconstant trigonometric polynomial of degree at most $\max\{r,s\}$. Since $F(n) = \lambda\mu(n)\sin(2\pi r u_n) + \eta\cos(2\pi s u_n)$, summing the three level-set errors gives

$$\mathfrak{D}_N \le 2\max\{r,s\} \sum_{z\in\{-1,0,1\}} \Delta_z(N). \tag{7.45}$$

Equations (7.42) and (7.43) prove the two asserted bounds. □

8. Statistical inference

This section concerns an explicitly random sample and is logically separate from the arithmetic sequence of Section 7. Throughout

$$X_1,\dots,X_N \overset{\text{i.i.d.}}{\sim} P^{r,s}_{\vartheta_0}, \qquad \vartheta_0 = (\lambda_0,\eta_0,q_0), \tag{8.1}$$

where the distinct coprime frequencies (r,s) are initially known. Write ϕ_ϑ and F_ϑ for the characteristic function and distribution function of $P^{r,s}_\vartheta$. The sampling statement (8.1) is part of every theorem in this section. A temporally correlated quasimomentum trajectory, a Markov phase process, and the deterministic arithmetic sequence do not satisfy it merely because their one-dimensional marginals match $P^{r,s}_\vartheta$. No block-bootstrap, mixing, or spectral-variance claim is made here.

8.1. **Regular parameter chambers.** Let $\mathcal{D}_{r,s}$ denote the finite set of positive amplitude ratios produced by Theorem 5.2. Fix a compact set

$$K \subset (0,\infty)^2 \times (0,1) \tag{8.2}$$

with $\vartheta_0 \in \operatorname{int} K$. We impose the following explicit chamber conditions. There is a $\delta > 0$ such that, for every $\vartheta = (\lambda,\eta,q) \in K$,

$$\operatorname{dist}\left(\frac{\lambda}{\eta}, \mathcal{D}_{r,s}\right) \ge \delta, \tag{8.3}$$

$$\operatorname{dist}\big(\{-\eta,\eta\}, \operatorname{CritVal}(h_\vartheta)\big) \ge \delta. \tag{8.4}$$

We also assume that $\vartheta \mapsto P^{r,s}_\vartheta$ is injective on K. This is a local assumption: Proposition 4.2 supplies such a set after shrinking around any locally identifiable interior point satisfying (8.4). No global identifiability assertion is used. Compactness gives constants

$$0 < \underline{\lambda} \le \lambda \le \overline{\lambda}, \quad 0 < \underline{\eta} \le \eta \le \overline{\eta}, \quad 0 < \underline{q} \le q \le \overline{q} < 1, \tag{8.5}$$

and hence $|X| \le B := \overline{\lambda} + \overline{\eta}$ uniformly on K.

8.2. **Moment and ECF estimators.** Choose integers $1 \le p_1 < \cdots < p_J$, $J \ge 3$, and put

$$z(x) = (x^{p_1},\dots,x^{p_J})^\top, \qquad m(\vartheta) = E_\vartheta z(X). \tag{8.6}$$

Every component of m is the finite resonance sum in Proposition 3.2. Let W be a fixed positive-definite $J \times J$ matrix and

$$\widehat{m}_N = \frac{1}{N}\sum_{i=1}^N z(X_i), \qquad Q_N^{\mathrm{mom}}(\vartheta) = \big(\widehat{m}_N - m(\vartheta)\big)^\top W \big(\widehat{m}_N - m(\vartheta)\big). \tag{8.7}$$

A moment estimator is any measurable minimizer

$$\widehat{\vartheta}_N^{\mathrm{mom}} \in \operatorname*{argmin}_{\vartheta\in K} Q_N^{\mathrm{mom}}(\vartheta). \tag{8.8}$$

A fixed measurable tie-breaking rule is used whenever the argmin is not a singleton. For this estimator we explicitly require that m be injective on K. Local identifiability of the full law alone does not imply that an arbitrarily selected finite list of moments is identifying.

For the ECF estimator, let $w \geq 0$ be integrable, supported in a compact interval $[-T, T]$, and positive almost everywhere on some nonempty open interval. Set

$$\widehat{\phi}_N(t) = \frac{1}{N}\sum_{i=1}^{N} e^{itX_i}, \qquad Q_N^{\mathrm{ecf}}(\vartheta) = \int_{-T}^{T} |\widehat{\phi}_N(t) - \phi_\vartheta(t)|^2 w(t)\, dt, \tag{8.9}$$

and define

$$\widehat{\vartheta}_N^{\mathrm{ecf}} \in \operatorname*{argmin}_{\vartheta \in K} Q_N^{\mathrm{ecf}}(\vartheta). \tag{8.10}$$

The same fixed measurable tie-breaking convention is used here. This is a continuum minimum-distance ECF estimator of the type introduced in [4, 3]. Both the real and imaginary parts are retained; discarding the imaginary part would lose the asymmetry information of Theorem 4.1 when s is even.

Theorem 8.1 (Existence and consistency)**.** *Assume* (8.1)–(8.5).

(i) *If the moment map m is injective on K, then the set in* (8.8) *is nonempty for every sample and every sequence of its minimizers satisfies*

$$\widehat{\vartheta}_N^{\mathrm{mom}} \longrightarrow \vartheta_0 \quad \textit{almost surely.} \tag{8.11}$$

(ii) *The set in* (8.10) *is nonempty for every sample and every sequence of its minimizers satisfies*

$$\widehat{\vartheta}_N^{\mathrm{ecf}} \longrightarrow \vartheta_0 \quad \textit{almost surely.} \tag{8.12}$$

Proof. The resonance formula shows that m is continuous. Hence Q_N^{mom} is continuous on the compact set K and attains its minimum. As a function of the sample and parameter it is measurable in the sample and continuous in the parameter. Its argmin correspondence therefore has nonempty compact values and a measurable graph; the measurable-selection theorem supplies the stipulated measurable minimizer. Since $|X| \leq B$, the strong law applied to the finite vector $z(X)$ gives $\widehat{m}_N \to m(\vartheta_0)$ almost surely. Consequently

$$\sup_{\vartheta \in K} \Big| Q_N^{\mathrm{mom}}(\vartheta) - \big(m(\vartheta_0) - m(\vartheta)\big)^{\top} W \big(m(\vartheta_0) - m(\vartheta)\big) \Big| \longrightarrow 0 \tag{8.13}$$

almost surely. The limiting objective is continuous and, by injectivity of m and positive definiteness of W, has the unique minimizer ϑ_0. The standard compactness argument is elementary: for every $\varepsilon > 0$, its minimum on the compact set $\{\vartheta \in K : \|\vartheta - \vartheta_0\| \geq \varepsilon\}$ is a strictly positive number; uniform convergence then forces every sample minimizer into the ε-ball for all sufficiently large N. This proves (i).

For (ii), continuity of $\phi_\vartheta(t)$ in (ϑ, t) follows either from Theorem 3.1 or directly by dominated convergence in the latent representation. Thus the ECF objective is continuous on K and has a minimizer. It is again a Carathéodory criterion, so the same measurable-selection argument supplies a measurable minimizer. Moreover,

$$|e^{itx} - e^{iux}| \leq B|t - u| \qquad (|x| \leq B). \tag{8.14}$$

A finite grid in $[-T, T]$, the scalar strong law at the grid points, and (8.14) imply

$$\sup_{|t| \leq T} |\widehat{\phi}_N(t) - \phi_{\vartheta_0}(t)| \longrightarrow 0 \quad \text{almost surely.} \tag{8.15}$$

Because all characteristic functions have modulus at most one, (8.15) gives uniform convergence on K of the ECF objective to

$$Q^{\mathrm{ecf}}(\vartheta) = \int_{-T}^{T} |\phi_{\vartheta_0}(t) - \phi_\vartheta(t)|^2 w(t)\, dt. \tag{8.16}$$

If this quantity is zero, the two characteristic functions agree almost everywhere on an open interval, hence everywhere on that interval by continuity. The bounded support makes each characteristic

function entire, so the identity theorem extends the equality to all real t. The laws are equal, and injectivity on K gives $\vartheta = \vartheta_0$. The same compactness argument proves (ii). □

8.3. **Asymptotic normality and sandwich covariance.** Let

$$D_0 = \left.\frac{\partial m(\vartheta)}{\partial \vartheta^\top}\right|_{\vartheta=\vartheta_0}, \qquad \Omega_0 = \mathrm{Var}_{\vartheta_0}(z(X)). \tag{8.17}$$

The finite resonance sums make m twice continuously differentiable on a neighborhood of K.

For the ECF estimator, write

$$d_0(t) = \left.\frac{\partial \phi_\vartheta(t)}{\partial \vartheta}\right|_{\vartheta=\vartheta_0} \in \mathbb{C}^3. \tag{8.18}$$

The components can be evaluated without differentiating the Bessel series:

$$\partial_\lambda \phi_\vartheta(t) = q\, E_\Theta\left[it \sin(r\Theta) e^{ith_\vartheta(\Theta)}\right], \tag{8.19}$$

$$\partial_\eta \phi_\vartheta(t) = (1-q) E_\Theta\left[it\cos(s\Theta) e^{it\eta\cos(s\Theta)}\right] + qE_\Theta\left[it\cos(s\Theta)e^{ith_\vartheta(\Theta)}\right], \tag{8.20}$$

$$\partial_q \phi_\vartheta(t) = E_\Theta e^{ith_\vartheta(\Theta)} - J_0(\eta t). \tag{8.21}$$

Define

$$A_0 = 2\,\mathrm{Re}\int_{-T}^{T} \overline{d_0(t)} d_0(t)^\top w(t)\, dt, \tag{8.22}$$

$$\zeta_0(x) = 2\,\mathrm{Re}\int_{-T}^{T} (e^{itx} - \phi_{\vartheta_0}(t))\overline{d_0(t)} w(t)\, dt, \tag{8.23}$$

$$B_0 = E_{\vartheta_0}[\zeta_0(X)\zeta_0(X)^\top]. \tag{8.24}$$

Theorem 8.2 (Asymptotic normality)**.** *Assume the hypotheses of Theorem 8.1 and $\vartheta_0 \in \mathrm{int}\, K$.*

(i) *If D_0 has column rank three, then*

$$\sqrt{N}(\widehat{\vartheta}_N^{\mathrm{mom}} - \vartheta_0) \Longrightarrow N_3(0, V_{\mathrm{mom}}), \tag{8.25}$$

where

$$V_{\mathrm{mom}} = (D_0^\top W D_0)^{-1} D_0^\top W \Omega_0 W D_0 (D_0^\top W D_0)^{-1}. \tag{8.26}$$

(ii) *If A_0 is positive definite, then*

$$\sqrt{N}(\widehat{\vartheta}_N^{\mathrm{ecf}} - \vartheta_0) \Longrightarrow N_3(0, V_{\mathrm{ecf}}), \tag{8.27}$$

with the ECF sandwich matrix

$$V_{\mathrm{ecf}} = A_0^{-1} B_0 A_0^{-1}. \tag{8.28}$$

Proof. For the moment estimator, consistency and interiority imply that, with probability tending to one, the minimizer is interior and satisfies the first-order equation. Taylor expansion of that equation around ϑ_0 gives

$$\sqrt{N}(\widehat{\vartheta}_N^{\mathrm{mom}} - \vartheta_0) = (D_0^\top W D_0)^{-1} D_0^\top W \frac{1}{\sqrt{N}} \sum_{i=1}^{N} (z(X_i) - m(\vartheta_0)) + o_P(1). \tag{8.29}$$

Indeed, the Hessian of the population criterion at ϑ_0 is $2D_0^\top W D_0$, while the terms containing second derivatives of m are multiplied by $\widehat{m}_N - m(\vartheta_0) = O_P(N^{-1/2})$ and are therefore negligible. The multivariate central limit theorem and Slutsky's theorem yield (8.25)–(8.26).

For the ECF criterion, differentiation under the integral is justified by compact t-support and the uniform bound $|X| \le B$. At ϑ_0 its gradient is

$$\nabla Q_N^{\mathrm{ecf}}(\vartheta_0) = -\frac{1}{N}\sum_{i=1}^{N} \zeta_0(X_i). \tag{8.30}$$

The population Hessian is A_0. The empirical Hessian at any random point converging to ϑ_0 converges in probability to A_0: its leading term is the integral in (8.22), and its remaining term contains $\widehat{\phi}_N - \phi_\vartheta$, which tends uniformly to zero. Hence the first-order equation and Taylor's theorem give

$$\sqrt{N}(\widehat{\vartheta}_N^{\mathrm{ecf}} - \vartheta_0) = A_0^{-1}\frac{1}{\sqrt{N}}\sum_{i=1}^{N} \zeta_0(X_i) + o_P(1). \tag{8.31}$$

The vector $\zeta_0(X)$ is bounded, has mean zero, and has covariance B_0. The multivariate central limit theorem proves (8.27)–(8.28). □

For computation, let $\widehat{D}$, $\widehat{\Omega}$, $\widehat{d}(t)$, $\widehat{A}$, and $\widehat{B}$ be the plug-in analogues at the relevant estimator, with

$$\widehat{\Omega} = \frac{1}{N}\sum_{i=1}^{N}(z(X_i) - \widehat{m}_N)(z(X_i) - \widehat{m}_N)^\top, \tag{8.32}$$

$$\widehat{A} = 2\,\mathrm{Re}\int_{-T}^{T} \overline{\widehat{d}(t)}\widehat{d}(t)^\top w(t)\,dt, \tag{8.33}$$

$$\widehat{\zeta}_i = 2\,\mathrm{Re}\int_{-T}^{T}(e^{itX_i} - \widehat{\phi}_N(t))\overline{\widehat{d}(t)}w(t)\,dt, \qquad \widehat{B} = \frac{1}{N}\sum_{i=1}^{N}\widehat{\zeta}_i\widehat{\zeta}_i^\top. \tag{8.34}$$

Define

$$\widehat{V}_{\mathrm{mom}} = (\widehat{D}^\top W\widehat{D})^{-1}\widehat{D}^\top W\widehat{\Omega}W\widehat{D}(\widehat{D}^\top W\widehat{D})^{-1}, \tag{8.35}$$

$$\widehat{V}_{\mathrm{ecf}} = \widehat{A}^{-1}\widehat{B}\widehat{A}^{-1}. \tag{8.36}$$

Proposition 8.3 (Consistency of the sandwich estimators)**.** *Under the hypotheses of the corresponding part of Theorem 8.2,*

$$\widehat{V}_{\mathrm{mom}} \longrightarrow_P V_{\mathrm{mom}}, \qquad \widehat{V}_{\mathrm{ecf}} \longrightarrow_P V_{\mathrm{ecf}}. \tag{8.37}$$

Proof. Consistency of the parameter estimators and continuity of the finite resonance sums give $\widehat{D} \to_P D_0$. Boundedness of $z(X)$ and the law of large numbers give $\widehat{\Omega} \to_P \Omega_0$. The rank condition keeps $\widehat{D}^\top W\widehat{D}$ invertible with probability tending to one, so continuity of matrix inversion proves the first convergence.

For the ECF estimator, differentiation under the phase integral is uniform on the compact set $K \times [-T,T]$; therefore $\sup_{|t|\le T}\|\widehat{d}(t) - d_0(t)\| \to_P 0$ and $\widehat{A} \to_P A_0$. Define $\zeta_\vartheta(x)$ by replacing ϑ_0 with ϑ in (8.23). Compact support of X and of w makes this class uniformly bounded and Lipschitz in ϑ. Consequently,

$$\frac{1}{N}\sum_{i=1}^{N}\zeta_{\widehat{\vartheta}_N}(X_i)\zeta_{\widehat{\vartheta}_N}(X_i)^\top \longrightarrow_P B_0. \tag{8.38}$$

Replacing $\phi_{\widehat{\vartheta}_N}$ in these scores by $\widehat{\phi}_N$ changes each score uniformly by $o_P(1)$ because of (8.15); hence $\widehat{B} \to_P B_0$. Positive definiteness of A_0 and continuity of inversion prove the second convergence. □

Corollary 8.4 (Asymptotic confidence intervals)**.** *Assume the corresponding covariance matrix V in Theorem 8.2 is positive definite and let $\widehat{V} \to V$ in probability. For $j \in \{1,2,3\}$,*

$$\left[\widehat{\vartheta}_{N,j} - z_{1-\alpha/2}\sqrt{\widehat{V}_{jj}/N}, \widehat{\vartheta}_{N,j} + z_{1-\alpha/2}\sqrt{\widehat{V}_{jj}/N}\right] \tag{8.39}$$

has coverage converging to $1-\alpha$. The ellipsoid

$$N(\vartheta - \widehat{\vartheta}_N)^\top \widehat{V}^{-1}(\vartheta - \widehat{\vartheta}_N) \le \chi^2_{3,1-\alpha} \tag{8.40}$$

has joint coverage converging to $1-\alpha$. These are asymptotic, not finite-sample, coverage statements.

Proof. Studentization in each coordinate and in the quadratic form follows from Theorem 8.2, consistency of $\widehat{V}$, and Slutsky's theorem. □

8.4. **Finite-sample confidence sets.** The continuous pushforward laws in the regular chamber permit distribution-free finite-sample inversion. Let F_N be the empirical distribution function and define

$$D_N(\vartheta) = \sup_x |F_N(x) - F_\vartheta(x)|. \tag{8.41}$$

Theorem 8.5 (Exact and conservative KS inversion)**.** *Let $c^{\mathrm{KS}}_{N,\alpha}$ be a finite-sample upper critical value for the one-sample Kolmogorov–Smirnov statistic under a continuous distribution, chosen so that its nonrejection probability is at least $1-\alpha$. Then*

$$\mathcal{C}^{\mathrm{KS}}_{N,\alpha} = \{\vartheta \in K : D_N(\vartheta) \le c^{\mathrm{KS}}_{N,\alpha}\} \tag{8.42}$$

has finite-sample coverage at least $1-\alpha$. Exact randomized critical values can make the coverage exactly $1-\alpha$. Without exact KS tables, the explicit set

$$\mathcal{C}^{\mathrm{DKW}}_{N,\alpha} = \left\{\vartheta \in K : D_N(\vartheta) \le \sqrt{\frac{\log(2/\alpha)}{2N}}\right\} \tag{8.43}$$

has finite-sample coverage at least $1-\alpha$. Projection of either joint set onto one coordinate gives a conservative finite-sample confidence interval for that coordinate.

Proof. For every fixed ϑ, F_ϑ is continuous because it is a mixture of continuous circle pushforwards. Therefore the distribution of $D_N(\vartheta)$ under $P^{r,s}_\vartheta$ is the distribution-free one-sample KS law. In particular, at the true parameter the event defining (8.42) has probability at least $1-\alpha$. The Dvoretzky–Kiefer–Wolfowitz–Massart inequality [10] gives

$$P_{\vartheta_0}\left(D_N(\vartheta_0) > \varepsilon\right) \le 2e^{-2N\varepsilon^2}, \tag{8.44}$$

which yields (8.43). Whenever the joint set contains ϑ_0, every coordinate projection contains the corresponding true coordinate, so projection cannot reduce coverage. □

If the latent gate labels are observed, an additional exact interval is available.

Proposition 8.6 (Observed-gate interval)**.** *Suppose (M_i, X_i), rather than only X_i, are observed under* (8.1)*. Then*

$$A_N = \sum_{i=1}^{N} \mathbf{1}_{\{M_i \neq 0\}} \sim \mathrm{Binomial}(N, q_0). \tag{8.45}$$

Inverting the two binomial tails gives the usual equal-tailed Clopper–Pearson interval for q_0, with finite-sample coverage at least $1-\alpha$.

Proof. The indicators in (8.45) are independent Bernoulli variables with success probability q_0. For each candidate q, retain it precisely when the observed count lies in the central binomial acceptance region whose two excluded tails have probability at most $\alpha/2$. At $q = q_0$ the probability of exclusion is at most the sum of those two tail probabilities, hence at most α. □

When only X_i is observed, the gate count is latent and this binomial pivot is not available. The KS inversion remains valid because it uses the full observed law.

8.5. **Parametric bootstrap.** For either regular estimator $\widehat{\vartheta}_N$, generate conditionally independent bootstrap variables by

$$M_i^* \sim (1-\widehat{q}_N, \widehat{q}_N/2, \widehat{q}_N/2), \quad \Theta_i^* \sim \mathrm{Unif}[0,2\pi), \quad X_i^* = \widehat{\lambda}_N M_i^* \sin(r\Theta_i^*) + \widehat{\eta}_N \cos(s\Theta_i^*), \tag{8.46}$$

then recompute the same estimator as $\widehat{\vartheta}_N^*$.

Theorem 8.7 (Validity of the parametric bootstrap)**.** *Assume the hypotheses of Theorem 8.2. In addition, assume that the relevant rank condition is uniform on a neighborhood U of ϑ_0: the smallest eigenvalue of $D(\vartheta)^\top W D(\vartheta)$ for the moment estimator, or of $A(\vartheta)$ for the ECF estimator, is bounded below by a positive constant on U. Then, conditionally on the data,*

$$\mathcal{L}^*\Big(\sqrt{N}(\widehat{\vartheta}_N^* - \widehat{\vartheta}_N)\Big) \Longrightarrow_P N_3(0, V), \tag{8.47}$$

where V is the same covariance matrix as in Theorem 8.2. Consequently, the basic bootstrap interval

$$\left[\widehat{\vartheta}_{N,j} - \frac{c^*_{j,1-\alpha/2}}{\sqrt{N}}, \widehat{\vartheta}_{N,j} - \frac{c^*_{j,\alpha/2}}{\sqrt{N}}\right], \tag{8.48}$$

where $c^*_{j,u}$ is the conditional u-quantile of $\sqrt{N}(\widehat{\vartheta}^*_{N,j} - \widehat{\vartheta}_{N,j})$, has coverage converging to $1-\alpha$ whenever $V_{jj} > 0$. Its asserted coverage is asymptotic.

Proof. By Theorem 8.1, $\widehat{\vartheta}_N \to \vartheta_0$ in probability, so with probability tending to one it lies in U. Conditional on the data, the bootstrap observations are i.i.d. from $P^{r,s}_{\widehat{\vartheta}_N}$ and remain uniformly bounded by a deterministic constant. For the moment estimator, the conditional central limit theorem gives

$$\frac{1}{\sqrt{N}} \sum_{i=1}^{N} \big(z(X_i^*) - m(\widehat{\vartheta}_N)\big) \Longrightarrow_P N_J(0, \Omega_0), \tag{8.49}$$

because boundedness makes the conditional Lindeberg condition automatic and $\Omega(\widehat{\vartheta}_N) \to \Omega_0$. To justify conditional consistency explicitly, choose a compact neighborhood $U_0 \subset U \cap \operatorname{int} K$ of ϑ_0. For every $\varepsilon > 0$, continuity, compactness, and injectivity give a uniform identification gap

$$c_\varepsilon = \inf_{\substack{\vartheta \in U_0,\ u \in K \\ \|u-\vartheta\| \geq \varepsilon}} Q_\vartheta(u) > 0, \tag{8.50}$$

where Q_ϑ is the population moment or ECF criterion when the generating parameter is ϑ. Conditional bounded laws of large numbers make the bootstrap criterion converge uniformly to $Q_{\widehat{\vartheta}_N}$ on K. Since $P(\widehat{\vartheta}_N \in U_0) \to 1$, the same separation argument used in Theorem 8.1 gives $\widehat{\vartheta}_N^* - \widehat{\vartheta}_N = o_{P^*}(1)$ in outer probability. The same Taylor expansion as in (8.29), now around $\widehat{\vartheta}_N$, therefore gives the conditional linear representation with all matrices evaluated at $\widehat{\vartheta}_N$. Their convergence to the matrices at ϑ_0 proves (8.47).

For the ECF estimator, use the bootstrap score obtained from (8.23) with parameter $\widehat{\vartheta}_N$. Its summands are conditionally i.i.d., centered, and uniformly bounded; their covariance converges to B_0. The conditional central limit theorem, conditional consistency, and the uniform positive lower bound on the ECF Hessian give the bootstrap counterpart of (8.31). This proves (8.47) for the ECF estimator. Convergence of the conditional distribution to a normal law with a continuous marginal distribution implies convergence of its conditional quantiles. Combining this with the unconditional limit in Theorem 8.2 proves the stated coverage of (8.48). □

The Van Hove exponents occurring here are $-1/2$, $-2/3$, and $-3/4$, all strictly greater than -1 and therefore locally integrable. More importantly, neither bootstrap proof uses a density likelihood or differentiates through a singular density: it uses bounded moments or bounded characteristic-function scores. The regular-chamber assumptions exclude the parameter surfaces on which critical-point topology changes. Thus the theorem covers singular DOS within a fixed regular chamber, but it does not assert nonregular validity at a cubic or quartic transition.

8.6. **Unknown frequencies as discrete model selection.** Frequency inference is not absorbed into the preceding continuous-parameter theory. Let $\mathcal{R}$ be a finite collection of distinct reduced pairs (r, s) with $r \neq s$. For each $a = (r, s) \in \mathcal{R}$, let K_a be a compact regular chamber and let $\phi_{a,\vartheta}$ be its characteristic function. Define

$$\widehat{Q}_{N,a} = \inf_{\vartheta \in K_a} \int_{-T}^{T} |\widehat{\phi}_N(t) - \phi_{a,\vartheta}(t)|^2 w(t)\, dt, \qquad \widehat{a}_N \in \operatorname*{argmin}_{a \in \mathcal{R}} \widehat{Q}_{N,a}. \tag{8.51}$$

Theorem 8.8 (Consistency under an explicit cross-model gap)**.** *Suppose the true pair $a_0 \in \mathcal{R}$ and true parameter $\vartheta_0 \in \operatorname{int} K_{a_0}$ satisfy*

$$\Delta := \min_{a \neq a_0} \inf_{\vartheta \in K_a} \int_{-T}^{T} |\phi_{a_0,\vartheta_0}(t) - \phi_{a,\vartheta}(t)|^2 w(t)\, dt > 0. \tag{8.52}$$

Then $P(\widehat{a}_N = a_0) \to 1$. Let $\widehat{\vartheta}_N^{\mathrm{sel}}$ be the parameter estimator in the selected chamber and let $\widehat{\vartheta}_N^{\mathrm{orc}}$ be the estimator computed in the true chamber. With a common deterministic tie-breaking rule,

$$P\Big(\widehat{\vartheta}_N^{\mathrm{sel}} = \widehat{\vartheta}_N^{\mathrm{orc}}\Big) \longrightarrow 1. \tag{8.53}$$

Consequently the selected estimator has the same unconditional weak limit as the oracle estimator whenever the latter satisfies Theorem 8.2.

Proof. The uniform ECF convergence used in Theorem 8.1 holds for every member of the finite collection $\mathcal{R}$, hence simultaneously. The minimized criterion for a_0 converges to zero, while every wrong minimized criterion converges to a number at least Δ. Therefore the selected pair equals a_0 with probability tending to one. On that event the selected and oracle estimators minimize the same criterion on the same chamber and, under the common tie-breaking rule, are equal. This proves (8.53). Two random vectors that differ with probability tending to zero have the same weak limit whenever either limit exists. □

If (8.52) fails, the data can at most select the equivalence class of pairs attaining zero population distance. The theorem does not assert global frequency identifiability, and common frequency factors, $q = 0$, zero amplitudes, and the reducible case $r = s$ must be removed before constructing $\mathcal{R}$.

The size of the gap is practically important even when it is positive. A deterministic population audit minimized the same complex-ECF criterion over every wrong chamber used in the simulation. Table 1 displays the closest wrong pair. The low-active-fraction law is nearly confounded with $(2, 1)$ on the selected frequency window: its gap is two orders of magnitude smaller than that of the baseline. The discrepancy in moments of orders one through four is also much smaller than the corresponding discrepancy through order six. Thus the calculation gives a concrete case in which low moments are weak discriminators while the wider ECF retains a positive cross-model gap.

8.7. **Random-index inference for a finite Möbius population.** The i.i.d. calibration above must not be applied to the deterministic Möbius sequence. Fix a finite population size L and deterministic values

$$y_k = \lambda \mu(k) \sin(r\omega k \log k) + \eta \cos(s\omega k \log k), \qquad 1 \leq k \leq L. \tag{8.54}$$

TABLE 1. Closest wrong reduced-frequency model in the deterministic ECF audit.

True chamber	wrong pair	ECF gap	max. moment gap 1:4	max. moment gap 1:6
low active fraction	$(2,1)$	2.0×10^{-6}	0.0132	0.0534
symmetric baseline	$(3,1)$	2.27×10^{-4}	0.0662	0.2686
below cubic ratio	$(4,1)$	9.07×10^{-4}	0.1245	0.3726
above cubic ratio	$(4,1)$	1.115×10^{-3}	0.1086	0.3738
asymmetric, even s	$(5,2)$	4.481×10^{-3}	0.3251	1.5865
high active fraction	$(3,1)$	6.257×10^{-3}	0.1569	0.5324

Draw indices $I_1, \dots, I_n$ uniformly without replacement from $\{1, \dots, L\}$. The only randomness in this subsection is the sampling design. Put

$$F_L(x) = \frac{1}{L} \sum_{k=1}^{L} \mathbf{1}_{\{y_k \le x\}}, \qquad \widehat{F}_{n,L}(x) = \frac{1}{n} \sum_{j=1}^{n} \mathbf{1}_{\{y_{I_j} \le x\}}. \tag{8.55}$$

Theorem 8.9 (Pointwise exact finite-population interval)**.** *Fix x and write $K_x = LF_L(x)$ and $H_x = n\widehat{F}_{n,L}(x)$. Conditional on the finite population,*

$$H_x \sim \text{Hypergeometric}(L, K_x, n). \tag{8.56}$$

For each $k \in \{0, \dots, L\}$, let $\mathcal{A}_k$ be an equal-tailed acceptance set for this hypergeometric law whose two excluded tails each have probability at most $\alpha/2$. Then

$$\mathcal{K}_{x,\alpha}(H_x) = \{k : H_x \in \mathcal{A}_k\} \tag{8.57}$$

has design coverage at least $1 - \alpha$ for K_x. Dividing its smallest and largest elements by L gives a pointwise conservative interval for $F_L(x)$.

Proof. There are exactly K_x population indices with $y_k \le x$. A simple random sample of n indices therefore contains a hypergeometric number of such indices, proving (8.56). At the true K_x, rejection occurs only in the two excluded tails of $\mathcal{A}_{K_x}$, whose total probability is at most α. Inverting the acceptance regions proves the claim. □

Theorem 8.10 (Serfling simultaneous band)**.** *For $1 \le n < L$, define*

$$\varepsilon_{n,L,\alpha} = \sqrt{\frac{1 - (n-1)/L}{2n} \log \frac{2(L+1)}{\alpha}}. \tag{8.58}$$

Then, conditional on the fixed finite population,

$$P_{\text{design}}\left(\sup_x |\widehat{F}_{n,L}(x) - F_L(x)| \le \varepsilon_{n,L,\alpha} \right) \ge 1 - \alpha. \tag{8.59}$$

For $n = L$, the supremum is identically zero.

Proof. For fixed x, the population values $\mathbf{1}_{\{y_k \le x\}}$ lie in $[0,1]$. Serfling's inequality [11] gives

$$P_{\text{design}}\left(|\widehat{F}_{n,L}(x) - F_L(x)| > \varepsilon \right) \le 2 \exp\left(-\frac{2n\varepsilon^2}{1 - (n-1)/L} \right). \tag{8.60}$$

As x varies, the two empirical distribution functions can change only at the at most L distinct population values, together with one exterior interval. A union bound over at most $L+1$ thresholds and substitution of (8.58) prove (8.59). If $n = L$, every population index is observed and the two distribution functions coincide. □

Corollary 8.11 (Serfling interval for a finite-population mean)**.** *For any fixed statistic g with range contained in $[a, b]$, the interval*

$$\frac{1}{n}\sum_{j=1}^{n} g(y_{I_j}) \;\pm\; (b-a)\sqrt{\frac{1-(n-1)/L}{2n}\log\frac{2}{\alpha}} \tag{8.61}$$

has design coverage at least $1-\alpha$ for $L^{-1}\sum_{k=1}^{L} g(y_k)$ when $n < L$; for $n = L$ the sample mean equals the population mean exactly.

Proof. Apply Serfling's inequality to the fixed finite population $g(y_1), \ldots, g(y_L)$, whose range length is at most $b-a$, and solve the resulting two-sided tail bound for its radius. The case $n = L$ is deterministic. □

These are finite-population design statements. They do not estimate the infinite arithmetic limit in Theorem 7.1. If F_∞ denotes the continuous limiting distribution function and

$$d_L = \sup_x |F_L(x) - F_\infty(x)|, \tag{8.62}$$

then the triangle inequality combines the design band with d_L:

$$\sup_x |\widehat{F}_{n,L}(x) - F_\infty(x)| \le \varepsilon_{n,L,\alpha} + d_L \tag{8.63}$$

with design probability at least $1-\alpha$. Theorem 7.2 supplies the deterministic bounds

$$d_L \ll_{A,\lambda,\eta,r,s,\omega} (\log L)^{-A} \tag{8.64}$$

for every fixed $A > 0$, with ineffective constants, and the effective alternative

$$d_L \ll_{\lambda,\eta,r,s,\omega} \frac{\log\log L}{(\log L)^{1/4}}. \tag{8.65}$$

Consequently, the right-hand side of (8.63) can be made fully quantitative by adding either deterministic rate. These asymptotic big-O bounds do not provide a numerically evaluated constant or threshold for the available finite file.

8.8. **Simulation audit.** The simulation study accompanying the source code examines bias and mean squared error, coverage for Wald intervals, the parametric bootstrap, and KS/DKW sets, discrete frequency selection, and finite-population Serfling coverage. It does so in several null configurations: symmetric and asymmetric laws, points on both sides of the $(r, s) = (2, 1)$ cubic ratio while retaining a positive separation, low and high active-gate fractions, and several competing reduced frequency pairs. Each simulated configuration is checked against the degeneracy-ratio list before sampling. The numerical ECF criterion uses deterministic quadrature and retains both its real and imaginary parts.

Table 2 reports a first reproducible audit with $R = 100$ independent samples in each chamber at $N = 1000$. The bias and RMSE columns take the maximum over (λ, η, q), and the Wald column gives the range of the three marginal empirical coverages. The final column gives exact-KS/DKW coverage. At a true coverage of 0.95, the Monte Carlo standard error based on $R = 100$ is approximately 0.022; consequently the table is a diagnostic rather than a precision coverage study. The committed data files also contain the corresponding $N = 250$ results and all parameterwise values.

The finite-population audit used $L = 100{,}000$, $R = 200$ random samples without replacement, four deterministic Möbius populations (baseline, asymmetric, and the two sides of the cubic ratio), and two sample sizes. Table 3 reports ranges over those four populations. The pointwise column ranges over three population quantiles as well as the four populations. The value 1.000 in the Serfling columns reflects the deliberate conservatism of those finite-population bounds in this experiment; it is not an assertion of exact unit coverage in general.

TABLE 2. Monte Carlo audit at $N = 1000$ with $R = 100$ replications per chamber.

Null chamber	estimator	max. |bias|	max. RMSE	Wald coverage	KS/DKW
asymmetric, even s	ECF	0.005	0.045	0.92–0.97	0.98/0.99
asymmetric, even s	moments	0.002	0.054	0.94–0.99	0.98/0.99
high q	ECF	0.006	0.032	0.95–0.95	0.96/0.97
high q	moments	0.008	0.049	0.93–0.96	0.96/0.97
low q	ECF	0.016	0.061	0.92–0.99	0.95/0.95
low q	moments	0.004	0.032	0.92–0.95	0.95/0.95
above cubic ratio	ECF	0.009	0.046	0.96–1.00	0.99/0.99
above cubic ratio	moments	0.003	0.033	0.97–1.00	0.99/0.99
below cubic ratio	ECF	0.007	0.077	1.00–1.00	0.95/0.95
below cubic ratio	moments	0.006	0.053	0.92–1.00	0.95/0.95
symmetric baseline	ECF	0.002	0.038	0.97–0.97	0.96/0.96
symmetric baseline	moments	0.002	0.037	0.94–0.96	0.96/0.96

TABLE 3. Finite-population design audit with $L = 100{,}000$ and $R = 200$.

n	Serfling CDF	Serfling mean	mean $\|\widehat{F} - F_L\|_\infty$	pointwise exact
1,000	1.000	1.000	0.0267–0.0273	0.940–0.980
10,000	1.000	1.000	0.0079–0.0087	0.925–0.980

These Monte Carlo results are numerical audits of the proved statements; they are not used in any proof and do not upgrade asymptotic coverage to finite-sample coverage.

9. Numerical audit of the arithmetic example

A chunked pass over 455,052,511 stored Möbius values gave

$$\begin{aligned} \text{variance:} &\quad 0.335227886 \text{ versus } 0.335213551, \\ \text{fourth moment:} &\quad 0.286443285 \text{ versus } 0.286430673, \\ \text{binned CDF discrepancy:} &\quad 3.4636 \times 10^{-5}. \end{aligned} \tag{9.1}$$

The input is a signed-`int8` file beginning with $\mu(1)$ and has SHA-256

$$\texttt{0e551f47d924ef6f3e348e131fdf53e0c677ad5e5471ef9d6249b3fd662e2ea4}. \tag{9.2}$$

Every stored byte was checked to lie in $\{-1, 0, 1\}$. The reported CDF number is the maximum difference of cumulative masses on 4096 equal bins spanning $[-1.25, 1.25]$; the support lies inside this interval, and the bins follow the `numpy.histogram` convention (half-open except for the final closed bin). The reference masses use eight million equally spaced phases and are normalized to sum to one. Raw moments were accumulated from 2,000,000-entry `float64` blocks with Kahan compensation between blocks. A second complete pass with 1,000,000-entry blocks reproduced the variance, fourth moment, histogram, and CDF number exactly at stored precision; the two means differed by 3.4×10^{-21}. Both machine-readable audit files and all prefix checkpoints are included in the supplement. Separately, 100,000 seeded indices were recomputed with 60-decimal phases. The maximum absolute value difference from the `float64` calculation was 3.72×10^{-6}; 29 assignments changed among the 4096 bins, and the two sampled binned CDFs differed by 1.0×10^{-5}. Sampled raw-moment differences were at most 1.5×10^{-9}. This is a sensitivity diagnostic rather than a deterministic error bound, so the CDF value in (9.1) is explicitly a result of the stated numerical protocol and is not a certified exact Kolmogorov discrepancy. A two-million-point sample

randomized within 200 equal strata gave 4.5330×10^{-4}. These numerical observations are not used to estimate the constants or thresholds in Theorem 7.2.

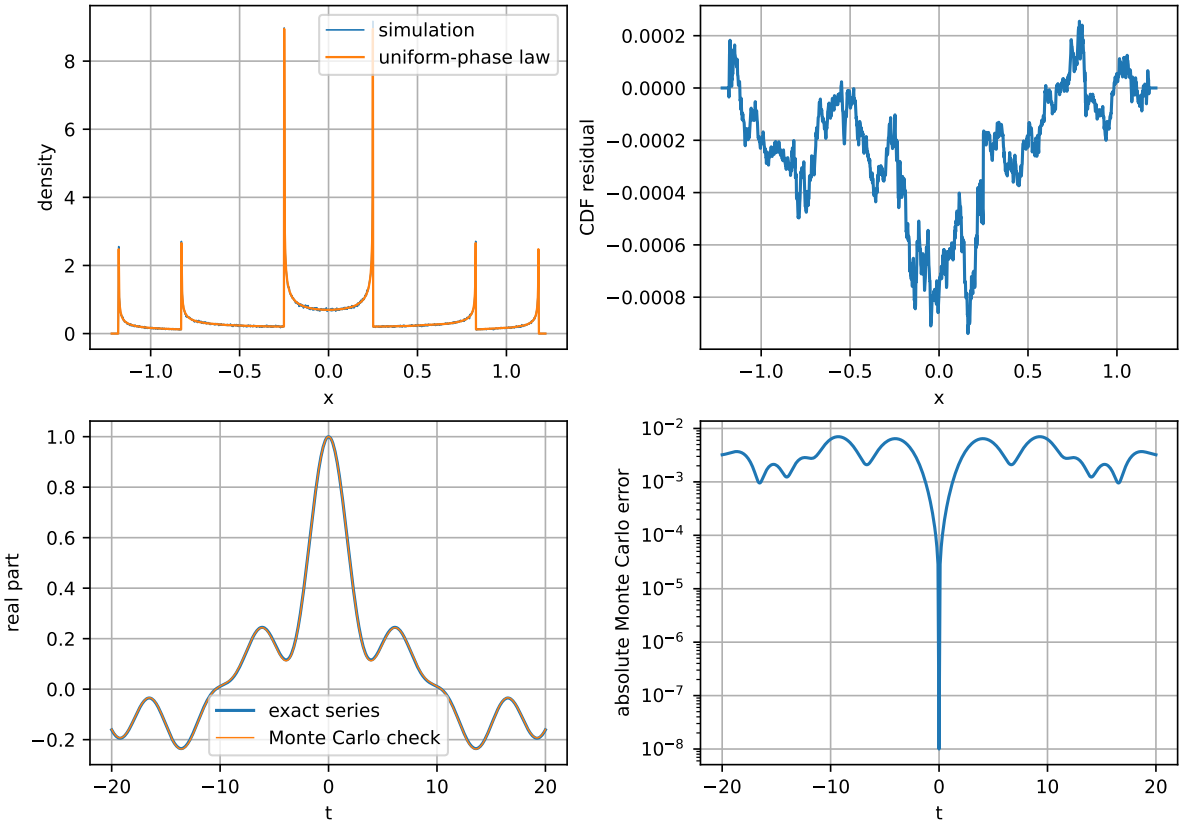


FIGURE 1. Density, CDF, and characteristic-function comparisons.

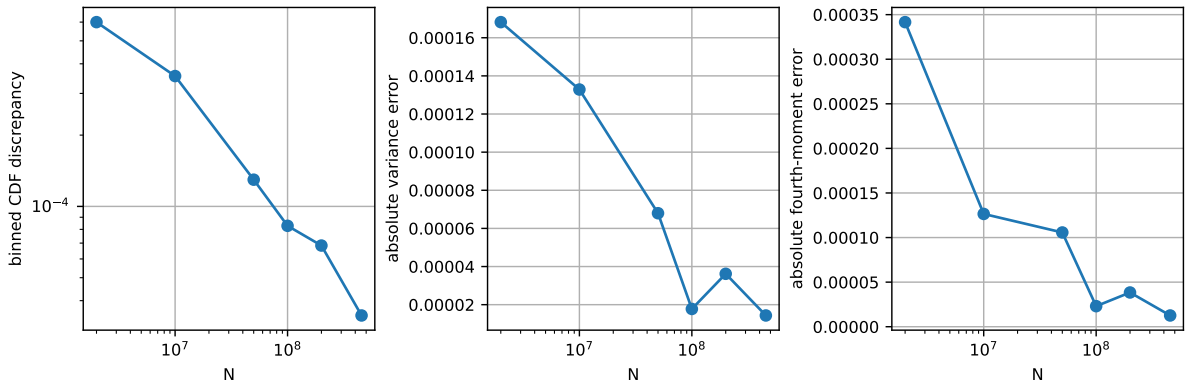


FIGURE 2. Streaming prefix diagnostics through the full file.

10. PHOTONIC REALIZATION: THE DUTT ET AL. EXPERIMENT

10.1. **Experimental input and scope.** Dutt et al. realized a synthetic frequency lattice in an electro-optically modulated fiber-ring resonator and measured its band structure directly in the

time domain [9]. Their platform supports reconfigurable long-range hopping and a photonic gauge phase. For the nonreciprocal experiment in their Fig. 4(c), the published drive is

$$V_M(t) = V_1 \cos(\Omega_R t) + V_2 \cos(2\Omega_R t + \phi), \qquad \frac{V_2}{V_1} = 0.40, \qquad \phi = \frac{\pi}{2}. \tag{10.1}$$

The experiment displayed a band strongly asymmetric about zero quasimomentum. The same work also showed, for nearest-neighbor hopping, that time-averaged transmission measures the photonic density of states (DOS), including finite-linewidth-broadened Van Hove edges.

We make a *published-parameter reconstruction*, not a fit to raw observations. The supporting raw data are not deposited with the article; its data-availability statement makes them available from the corresponding author on reasonable request. In particular, Fig. 4 contains no digitized experimental points. It states exactly which consequences follow from the reported harmonic drive and which curves are predictions of the enlarged family.

10.2. **Exact map into the harmonic family.** In the effective tight-binding description, write the two-range dispersion, in units of the first-harmonic energy scale A, as

$$\frac{\varepsilon_\phi(k)}{A} = \cos k + \alpha \cos(2k + \phi), \qquad \alpha = \frac{|J_2|}{|J_1|}. \tag{10.2}$$

At $\phi = \pi/2$, reflection $k \mapsto -k$ turns $\cos k - \alpha \sin(2k)$ into

$$h_\alpha(k) = \cos k + \alpha \sin(2k). \tag{10.3}$$

Consequently, the uniform-quasimomentum DOS in this model is exactly the $q = 1$ member of (1.3) with

$$(\lambda, \eta; r, s) = (\alpha, 1; 2, 1), \tag{10.4}$$

up to the overall energy scale A. Identifying the reported voltage ratio with $\alpha = 0.40$ uses the linear modulation-to-hopping calibration of the effective two-harmonic model; if calibrated hopping amplitudes differ from the voltage ratio, the same formulas apply after replacing 0.40 by $|J_2/J_1|$.

Experimental object	Harmonic-family object	Mathematical role
$k = \Omega_R t$ modulo 2π	Θ	uniform quasimomentum
nearest-neighbor hopping	$\eta \cos(s\Theta)$	$s = 1$, energy scale A
second-neighbor hopping	$\lambda \sin(r\Theta)$	$r = 2$, ratio $\lambda/\eta = \alpha$
gauge phases $\pm\pi/2$	$M = \mp 1$	time-reversed homogeneous domains
inactive second harmonic	$M = 0$	cosine/arcsine domain

TABLE 4. Dictionary between the photonic effective model and the probabilistic family. The last three rows describe a possible ensemble protocol, not sitewise disorder and not a claim about the single device in [9].

10.3. **Calibration, broadening, and active-fraction recovery.** The following stability statement quantifies the scope of the published-parameter reconstruction. Let $d_{\mathbb{T}}$ denote circular distance modulo 2π and let W_1 be the 1-Wasserstein distance.

Theorem 10.1 (Photonic calibration and broadening stability)**.** *For $K \sim \mathrm{Unif}[0, 2\pi)$, define*

$$E_{\alpha,\phi} = \cos K + \alpha \cos(2K + \phi), \qquad \alpha \geq 0. \tag{10.5}$$

Then

$$W_1(\mathcal{L}(E_{\alpha,\phi}), \mathcal{L}(E_{\alpha_0,\phi_0})) \leq |\alpha - \alpha_0| + \alpha_0 d_{\mathbb{T}}(\phi, \phi_0). \tag{10.6}$$

The same right-hand side bounds the Hausdorff distance between the two band supports. If an independent measurement error Z is added, then

$$W_1(\mathcal{L}(E_{\alpha,\phi}+Z),\mathcal{L}(E_{\alpha_0,\phi_0}))\le|\alpha-\alpha_0|+\alpha_0 d_{\mathbb{T}}(\phi,\phi_0)+\mathbb{E}|Z|. \tag{10.7}$$

For $Z\sim N(0,\sigma^2)$, the last term equals $\sigma\sqrt{2/\pi}$.

Proof. Couple both bands with the same K. The elementary inequality $|\cos u-\cos v|\le d_{\mathbb{T}}(u,v)$ gives

$$|E_{\alpha,\phi}-E_{\alpha_0,\phi_0}|\le|\alpha-\alpha_0|+alpha_0 d_{\mathbb{T}}(\phi,\phi_0). \tag{10.8}$$

Taking expectations proves (10.6). For any two continuous functions on the same compact domain, a uniform bound on their difference bounds the Hausdorff distance between their images, proving the support assertion. Couple the noisy variable by adding Z only to the first component and apply the triangle inequality to obtain (10.7). The Gaussian absolute first moment is $\sigma\sqrt{2/\pi}$. □

The active-domain fraction remains estimable after calibrated Gaussian broadening. Let

$$C_0(t)=J_0(t),\qquad C_{\alpha,\phi}(t)=\frac{1}{2\pi}\int_0^{2\pi}e^{it\{\cos k+\alpha\cos(2k+\phi)\}}\,dk, qquad\Delta(t)=C_{\alpha,\phi}(t)-C_0(t). \tag{10.9}$$

Theorem 10.2 (Exact one-frequency estimator under Gaussian broadening)**.** *Let $X_1,\dots,X_N$ be i.i.d. from the homogeneous-domain mixture that selects* $\cos K$ *with probability $1-q$ and $E_{\alpha,\phi}$ with probability q. Observe $Y_j=X_j+Z_j$, where the Z_j are independent $N(0,\sigma^2)$ variables and α,ϕ,σ are known. For a fixed $t\neq0$ with $\Delta(t)\neq0$, put*

$$\widehat{q}_N(t)=\frac{\mathrm{Re}\Big[\{e^{\sigma^2t^2/2}\widehat{\phi}_{Y,N}(t)-C_0(t)\}\overline{\Delta(t)}\Big]}{|\Delta(t)|^2}. \tag{10.10}$$

Then $\mathbb{E}\widehat{q}_N(t)=q$, $\widehat{q}_N(t)\to q$ almost surely, and

$$\sqrt{N}\{\widehat{q}_N(t)-q\}\Longrightarrow N\left(0,\mathrm{Var}\left[\frac{\mathrm{Re}\{e^{\sigma^2t^2/2}e^{itY_1}\overline{\Delta(t)}\}}{|\Delta(t)|^2}\right]\right). \tag{10.11}$$

Proof. Independence and the Gaussian characteristic function give

$$e^{\sigma^2t^2/2}\mathbb{E}e^{itY_1}=C_0(t)+q\Delta(t). \tag{10.12}$$

Substitution into (10.10) proves unbiasedness. The strong law applied to the bounded complex variables e^{itY_j} proves almost-sure consistency. Finally, (10.10) is an affine function of the sample mean of bounded real variables. The ordinary central limit theorem gives (10.11). □

A seeded audit with $q=0.65$, $N=2000$, and 300 replications confirms the distinction between known Gaussian broadening and hopping miscalibration. At measurement width $\sigma/A=0.05$, the mean estimates of q were 0.916, 0.651, and 0.509 when the nominal hopping ratios were respectively 0.32, 0.40, and 0.48, while the true ratio was 0.40. Thus the analytic deconvolution removes known Gaussian attenuation, but a 20% hopping error produces substantial model bias. This is precisely why a future comparison with raw spectra must calibrate $|J_2/J_1|$ rather than identify it automatically with a voltage ratio.

10.4. **A nearby, testable singularity transition.** The published ratio lies close to a sharp change in critical-point topology.

Proposition 10.3. *For h_α in* (10.3)*, let $\alpha>0$. If $0<\alpha<1/2$, there are two nondegenerate critical points per period. At $\alpha=1/2$, a cubic stationary point occurs at $k=3\pi/2$ and produces a DOS singularity proportional to $|x|^{-2/3}$. If $\alpha>1/2$, this point splits into two additional nondegenerate critical points, producing an additional pair of critical values.*

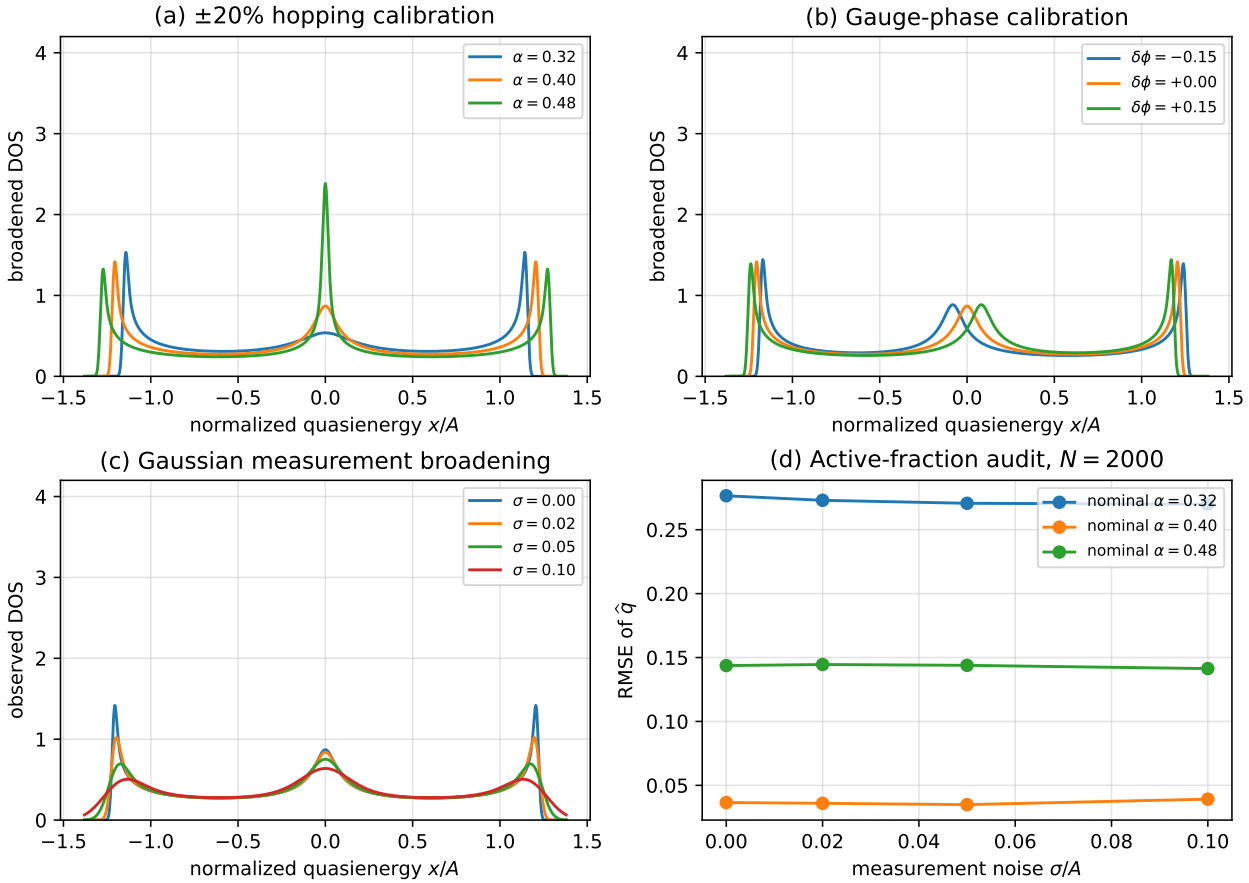


FIGURE 3. Sensitivity of the Dutt published-parameter reconstruction. (a) DOS under a ±20% hopping-ratio change. (b) Gauge-phase offsets. (c) Gaussian measurement broadening. (d) RMSE of the exact one-frequency active-fraction estimator under correct and misspecified hopping calibration. All curves are model calculations, not fits to experimental raw data.

Proof. With $u = \sin k$, the stationary equation is

$$4\alpha u^2 + u - 2\alpha = 0, \qquad u_\pm = \frac{-1 \pm \sqrt{1 + 32\alpha^2}}{8\alpha}. \tag{10.13}$$

The root u_+ is always admissible, whereas u_- enters $[-1, 1]$ precisely at $\alpha = 1/2$, when $u_- = -1$. Direct differentiation gives $h'_{1/2}(3\pi/2) = h''_{1/2}(3\pi/2) = 0$ and $h'''_{1/2}(3\pi/2) \neq 0$. Theorem 5.1 supplies the exponent. □

Proposition 10.4 (Quantitative unfolding of the cubic transition)**.** *Put $\alpha = 1/2 + \delta$ with $\delta > 0$. The cubic phase $3\pi/2$ splits into two ordinary critical phases*

$$k_\pm = \frac{3\pi}{2} \pm \frac{2}{\sqrt{3}}\delta^{1/2} + O(\delta^{3/2}). \tag{10.14}$$

Their critical values and curvatures satisfy

$$h_\alpha(k_\pm) = \mp\frac{8}{3\sqrt{3}}\delta^{3/2} + O(\delta^{5/2}), \tag{10.15}$$

$$|h''_\alpha(k_\pm)| = 2\sqrt{3}\,\delta^{1/2} + O(\delta^{3/2}). \tag{10.16}$$

Consequently, the separation of the two new singular energies is

$$\frac{16}{3\sqrt{3}}\delta^{3/2} + O(\delta^{5/2}), \tag{10.17}$$

and the local quadratic-DOS prefactor at either new phase is

$$\frac{q}{2\pi\, 3^{1/4}}\delta^{-1/4}\{1 + O(\delta)\}. \tag{10.18}$$

Proof. Write $k = 3\pi/2 + u$. Uniformly for small u and δ,

$$h_{1/2+\delta}(3\pi/2 + u) = -2\delta u + \left(\frac{1}{2} + \frac{4}{3}\delta\right) u^3 + O(u^5 + \delta u^5). \tag{10.19}$$

Differentiation and solution of the two nonzero stationary branches give $u_\pm = \pm 2\delta^{1/2}/\sqrt{3} + O(\delta^{3/2})$, proving (10.14). Substitution into (10.19) proves (10.15); differentiating twice proves (10.16). Subtraction gives (10.17). Finally, Theorem 5.1 with $m = 2$, $a_2 = h''_\alpha(k_\pm)/2$, and $N_2 = 2$ gives the prefactor $q(2/|h''_\alpha(k_\pm)|)^{1/2}/(2\pi)$, which is (10.18). □

Equation (10.17) supplies an explicit resolution scale. If the dimensionless linewidth is Γ/A, resolving the two new peaks requires it to be small relative to $16\delta^{3/2}/(3\sqrt{3})$; the statement is a scale requirement derived from the band, not a claim about the linewidth achieved in [9]. At the reported value $\alpha = 0.40$, the active-domain band therefore has only its two ordinary edge critical values,

$$x_{\text{edge}}/A = \pm 1.2146735483. \tag{10.20}$$

The critical ratio 0.50 is a prospective experimental target: tuning through it should create a central higher-order DOS anomaly and then split it. This statement is a model prediction, not a reported observation in [9].

10.5. **What the generalization adds.** The fixed-gauge experiment corresponds to a single homogeneous branch. The ternary gate becomes physical for an ensemble of independently prepared homogeneous domains:

$$\frac{\varepsilon_M(k)}{A} = \cos k + \alpha M \sin(2k), \qquad P(M = 0) = 1 - q, \quad P(M = \pm 1) = q/2. \tag{10.21}$$

Here $M = \pm 1$ represents balanced opposite gauge phases and $M = 0$ an inactive long-range drive. The ensemble DOS is then exactly

$$f_q(x) = (1 - q) f_{\cos}(x) + q f_{\text{active}}(x), \tag{10.22}$$

so q is the active-domain fraction. This extension gives simultaneously an analytic DOS, its critical exponents, and a topology diagram in α, with q interpreted as the active-domain fraction. These are model consequences of (10.21); they are not observations reported in [9]. The construction does not assert that the device has random sitewise hopping. An operational homogeneous-domain protocol would repeat a full band acquisition with one global drive setting held fixed during each acquisition: $+\pi/2$, $-\pi/2$, or second-harmonic off. Pooling complete acquisitions in proportions $q/2, q/2, 1 - q$ produces (10.22). It differs from sitewise disorder, which would randomize hopping from site to site within a single acquisition and would not be described by the mixture identity (10.22).

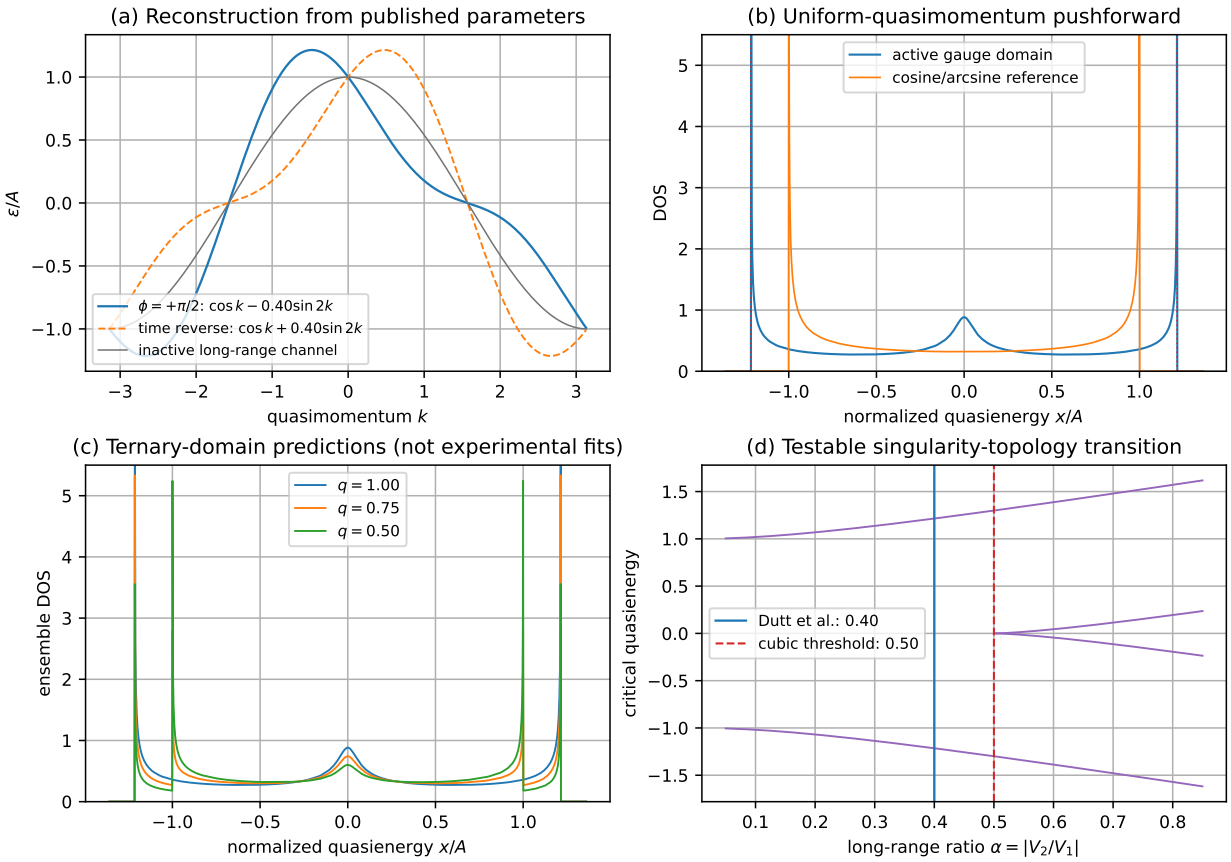


FIGURE 4. Application to the published parameters of Dutt et al. (a) The $\phi = \pi/2$ band and its time reverse; their pushforward laws coincide. (b) Phase-grid evaluation of the corresponding DOS and the inactive cosine reference. (c) Predictions for homogeneous-domain ensembles; these are not experimental fits. (d) Critical values versus second-neighbor ratio, showing the experimental value 0.40 and the predicted cubic threshold 0.50.

11. Homogeneous-domain interpretation

For a homogeneous domain indexed by M, take range-s hopping $t_s = \eta/2$ and range-r hopping $t_r = -i\lambda M/2$. The Hermitian Bloch dispersion is

$$\varepsilon_M(k) = t_s e^{isk} + \bar{t}_s e^{-isk} + t_r e^{irk} + \bar{t}_r e^{-irk} = \eta \cos(sk) + \lambda M \sin(rk). \tag{11.1}$$

Thus (1.3) is an ensemble of homogeneous domains with uniform quasimomentum. It is not sitewise disorder. The $M = \pm 1$ domains are conjugate and have the same DOS. The imaginary hopping is a Hermitian Peierls phase, not by itself non-Hermitian nonreciprocity. Photonic or Floquet relevance requires a platform-specific derivation.

References

[1] V. Bergelson, J. Kułaga-Przymus, M. Lemańczyk, and F. K. Richter, A generalization of Kátai's orthogonality criterion with applications, *Discrete Contin. Dyn. Syst.* 39 (2019), no. 5, 2581–2612, doi:10.3934/dcds.2019108.

[2] L. Classen and J. J. Betouras, High-order Van Hove singularities and their connection to flat bands, *Annu. Rev. Condens. Matter Phys.* 16 (2025), 229–251, doi:10.1146/annurev-conmatphys-042924-015000, arXiv:2405.20226.

[3] M. Carrasco and J.-P. Florens, Generalization of GMM to a continuum of moment conditions, *Econometric Theory* 16 (2000), no. 6, 797–834, doi:10.1017/S0266466600166010.

[4] A. Feuerverger and P. McDunnough, On the efficiency of empirical characteristic function procedures, *J. Roy. Statist. Soc. Ser. B* 43 (1981), no. 1, 20–27.
[5] S. W. Graham and G. Kolesnik, *Van der Corput's Method of Exponential Sums*, London Mathematical Society Lecture Note Series 126, Cambridge University Press, Cambridge, 1991, doi:10.1017/CBO9780511661976.
[6] L. Kuipers and H. Niederreiter, *Uniform Distribution of Sequences*, Wiley-Interscience, New York, 1974.
[7] K. Matomäki, X. Shao, T. Tao, and J. Teräväinen, Higher uniformity of arithmetic functions in short intervals I. All intervals, *Forum Math. Pi* 11 (2023), e29, doi:10.1017/fmp.2023.28, arXiv:2204.03754v4.
[8] NIST Digital Library of Mathematical Functions, Chapter 10, `https://dlmf.nist.gov/10`, accessed 2026-07-13.
[9] A. Dutt, M. Minkov, Q. Lin, L. Yuan, D. A. B. Miller, and S. Fan, Experimental band structure spectroscopy along a synthetic dimension, *Nature Communications* 10 (2019), Article 3122, doi:10.1038/s41467-019-11117-9.
[10] P. Massart, The tight constant in the Dvoretzky–Kiefer–Wolfowitz inequality, *Ann. Probab.* 18 (1990), no. 3, 1269–1283, doi:10.1214/aop/1176990746.
[11] R. J. Serfling, Probability inequalities for the sum in sampling without replacement, *Ann. Statist.* 2 (1974), no. 1, 39–48, doi:10.1214/aos/1176342611.
[12] L. Van Hove, The occurrence of singularities in the elastic frequency distribution of a crystal, *Phys. Rev.* 89 (1953), 1189–1193, doi:10.1103/PhysRev.89.1189.